Questions

Q1.

Figure 7 shows a box at rest on a floor.

The force that the floor exerts on the box is shown by the vector in Figure 7.

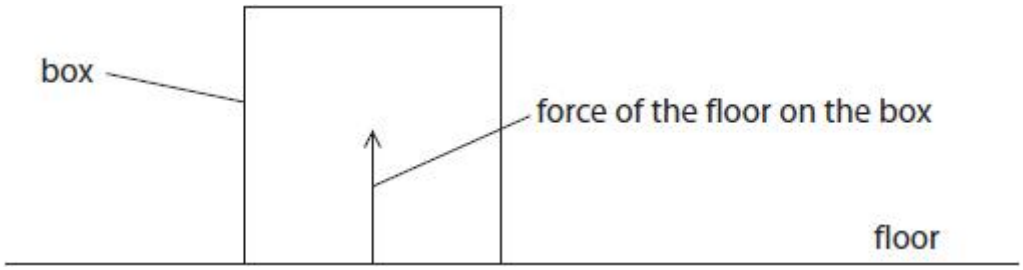


Figure 7

Add another vector to the diagram in Figure 7 to show the weight of the box.

(2)

(Total for question = 2 marks)

Q2.

A 20 cm ruler has a pivot at its centre.

The ruler is balanced when no forces act on the ruler.

Figure 4 shows two forces, Y and Z, acting on the ruler.

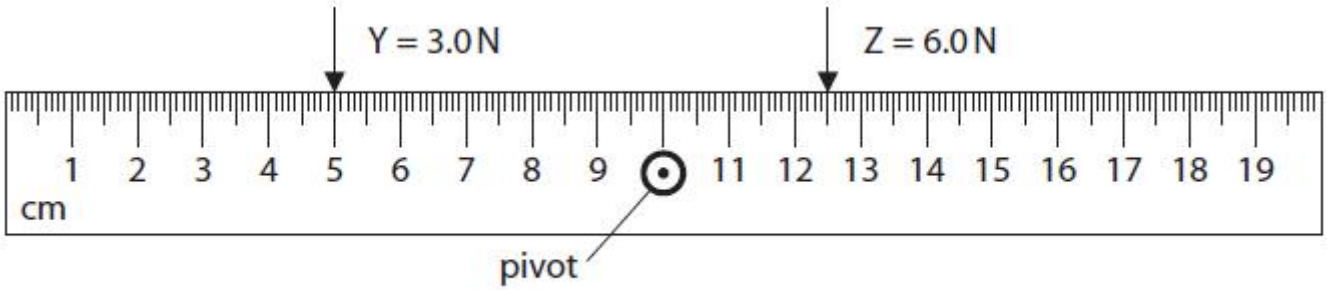


Figure 4

Use the principle of moments to show that the ruler in Figure 4 is balanced.

(2)

.....

.....

.....

.....

(Total for question = 2 marks)

Q3.

(i) Figure 9 shows two gears.

Gear Q moves clockwise as shown by the arrow.

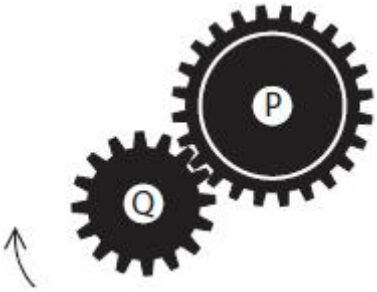


Figure 9

Which of these describes how gear P moves?

(1)

- ☐ **A** anticlockwise, slower than gear Q
- ☐ **B** anticlockwise, faster than gear Q
- ☐ **C** clockwise, slower than gear Q
- ☐ **D** clockwise, faster than gear Q

(ii) Figure 10 shows the number of teeth on the gears.

Gear P has 24 teeth, gear Q has 16 teeth.

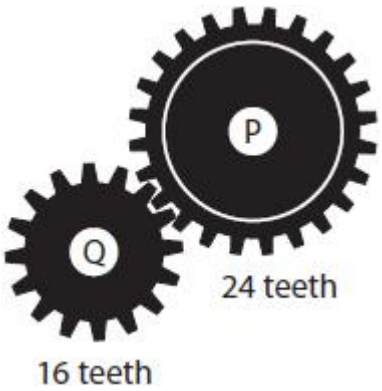


Figure 10

Which of these gives the ratio of the number of teeth on gear P to the number of teeth on gear Q?

(1)

- ☐ **A** 24 − 16
- ☐ **B** 16 + 24
- ☐ **C** 3 : 2
- ☐ **D** 2 : 3

(Total for question = 2 marks)

Q4.

Figure 8 shows part of a cart.

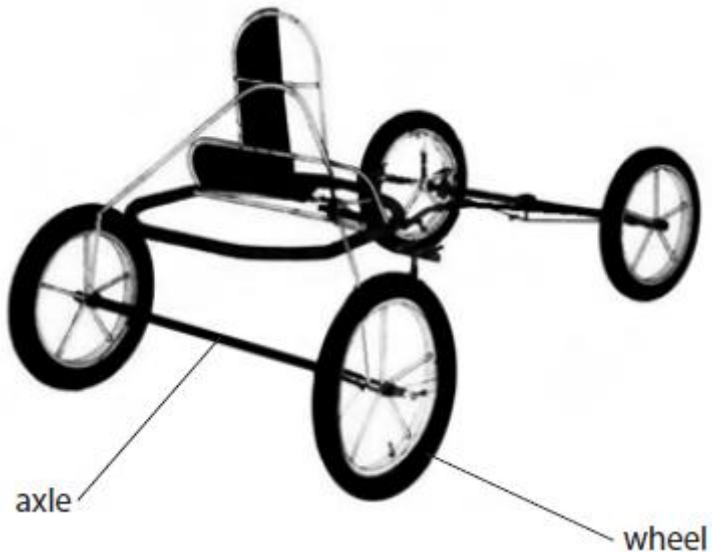


Figure 8

When the wheels turn the axles become warm.

(i) Explain why the axles become warm when the wheels turn.

(2)

.....

.....

.....

.....

(ii) Give **one** way of reducing the heating of the axles when the wheels turn.

(1)

.....

.....

(Total for question = 3 marks)

Q5.

Figure 5 shows an arrangement of three gears, P, Q and R.

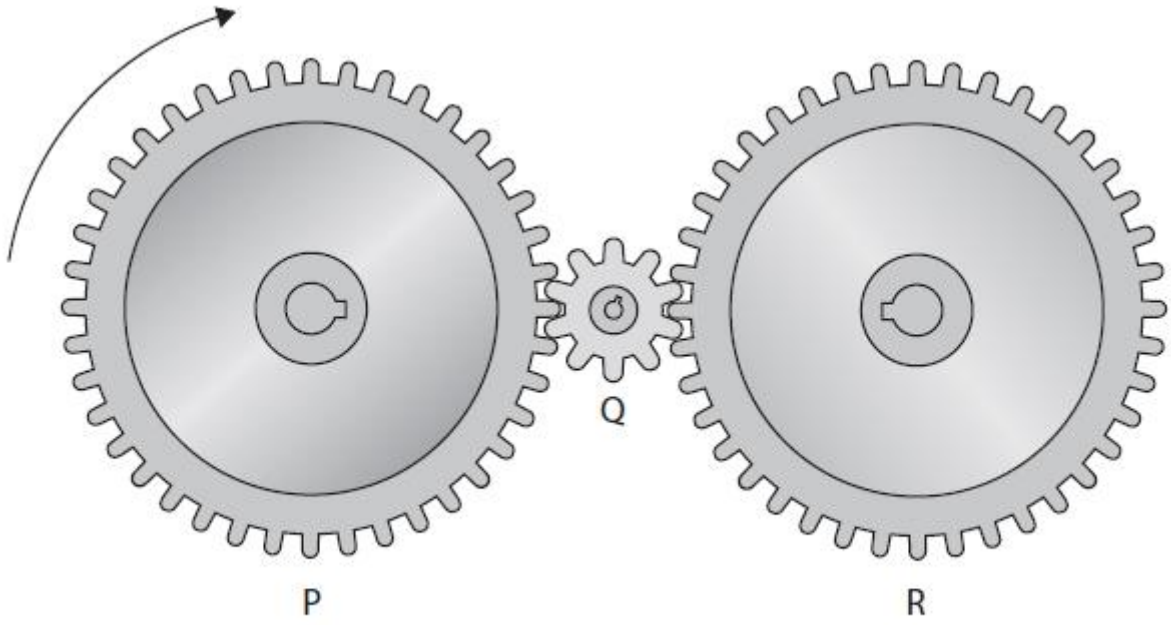


Figure 5

P and R have the same number of teeth.

P rotates in the direction shown.

State how the speed **and** direction of rotation of R compare with the speed and direction of rotation of P.

(2)

speed

.....

.....

direction

.....

.....

(Total for question = 2 marks)

Q6.

Figure 11 shows a lever used to lift a heavy load.

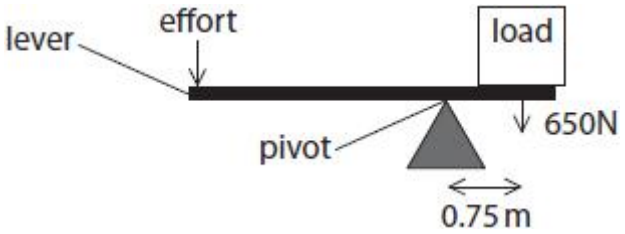


Figure 11

(i) The weight of the load is 650 N.

The centre of the load is 0.75 m from the pivot.
Calculate the moment of the load about the pivot.
State the unit.
Use the equation
moment = force × distance from the pivot

(3)

moment = ... unit .....

(ii) State the principle of moments.

(1)

.....

.....

(iii) An effort of 160 N is applied to the end of the lever to balance the load in Figure 11.

Calculate the distance between the effort and the pivot.

(3)

distance = ... m

(Total for question = 7 marks)

Q7.

Which of these is a vector?

(1)

- ☐ **A** distance
- ☐ **B** force
- ☐ **C** mass
- ☐ **D** work done

(Total for question = 1 mark)

Q8.

Which of these statements is true for a vector quantity?

(1)

☐ **A** It has size only

☐ **B** It has direction only

☐ **C** It has direction and size

☐ **D** It does not have direction or size

(Total for question = 1 mark)

Q9.

Figure 2 shows four items of equipment that can be used in investigations.

In which of these does a force cause rotation?

(1)

spring balance ☐ **A**

plotting compass ☐ **B**

thermometer ☐ **C**

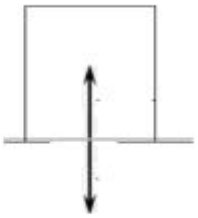
variable resistor ☐ **D**

Figure 2

(Total for question = 1 mark)

Mark Scheme

Q1.

Question Number:	Answer	Additional guidance	Mark
	 arrowed line vertically downwards (anywhere) (1) same length as vertical arrow upwards (1)	more than one line drawn 1 mark maximum judge by eye	**(2)** AO 1 1

Q2.

Question number	Answer	Additional guidance	Mark
	identification of clockwise and anticlockwise moment (1) 3(.0) x 5(.0) (/100) 6.0 x 2.5 (/100) values (of both moments) are equal (1)	15 and 15 seen or 0.15 and 0.15 seen Accept Y is half the force (as Z) but twice the distance (from the pivot as Z) for this mark may be implied by = sign (turning effect) of the two forces are equal	**(2)** **AO2**

Q3.

Question Number:	Answer	Mark
(i)	A anticlockwise, slower than gear Q **The only correct answer is A** ***B*** *is not correct as P is the larger gear and can only move slower than gear Q and anticlockwise* ***C*** *is not correct as gear P must be moving anticlockwise as gear Q is moving clockwise* ***D*** *is not correct as gear P must be moving anticlockwise as gear Q is moving clockwise*	**(1)** AO 3 2a

Question Number:	Answer	Mark
(ii)	C 3:2 **The only correct answer is C** ***A*** *is not correct as it is a subtraction* ***B*** *is not correct as it is an addition* ***D*** *is not correct as it gives the ratio of teeth on Q to teeth on P*	**(1)** AO 1 2

Q4.

Question Number:	Answer	Additional guidance	Mark
(i)	an explanation linking: wheel rubs on axle (as it rotates) OR friction (between the wheel and the axle) (1) causes heating/transfer of (thermal) energy/ work being done (1)	 allow generates heat	**(2)** AO 1 1

Question Number:	Answer	Additional guidance	Mark
(ii)	any one from: lubrication/oil (1) (ball) bearings / ball-race (1) go slower (1)	 anything that lubricates – grease etc.	(1) AO 1 1

Q5.

Question number	Answer	Additional guidance	Mark
	(speed of R is) same (as speed of) P (1) (sense/direction of R is) same (as sense/direction of P) (1)	 clockwise / to the right mark may be awarded by arrow on diagram (provided it is not contradicted by a statement)	(2) AO1

Q6.

Question Number:	Answer	Additional guidance	Mark
(i)	substitution(1) (moment) =650 x 0.75 evaluation(1) 490 unit (1) Nm	 accept any value that rounds to 490 e.g. 487.5 allow a maximum of 1 mark out of the first two marking points for a power of ten error independent mark award full marks for the correct answer without any working	**(3)** AO 1 1 AO 2 1

Question Number:	Answer	Additional guidance	Mark
(ii)	(sum of the) clockwise moments (about a point) = (sum of the) anticlockwise moments (about that point) (1)	idea that moments on each side of a pivot can be balanced	**(1)** AO 1 1

Question Number:	Answer	Additional guidance	Mark
(iii)		substitution and rearrangement in either order	(1) AO 2 1
	substitution(1) 160 x distance of effort from pivot = 490	accept 160 x distance of effort from pivot = 487.5 160 x distance from pivot =650 x 0.75	
	rearrangement (1) distance of effort from pivot = $\frac{490}{160}$	accept $\frac{650 \times 0.75}{160}$ $\frac{487.5}{160}$	
	evaluation (1) 3.1(m)	accept any value which rounds to 3 maximum of two marks for a power of ten error award full marks for the correct answer without working	

Q7.

Question number	Answer	Mark
	B force A, C and D are all scalars; B is the only vector	**(1)** **AO1**

Q8.

Question number	Answer	Mark
	C It has direction and size Option C is the only correct combination for a vector quantity	(1) AO1

Q9.

Question number	Answer	Mark
	B plotting compass A is incorrect because a force causes a linear movement. C is incorrect because the liquid column expands or contracts linearly. D is incorrect because the slider is moved linearly	(1) AO1